

Basic integrals

I (Df and DF)	f(x) (f is the given function)	f(x) (F is a function of f of a primitive number)	I (Df and DF)	f(x) (f is the given function)	f(x) (F is a function of f of a primitive number)
R	x^n ($n = 0, 1, 2, \dots$)	$\frac{x^{n+1}}{n+1}$	R	sh x	ch x
			R	ch x	sh x
(0, +∞)	$\frac{1}{x}$	ln x	R	th x	ln ch x
(-∞, 0)	$\frac{1}{x}$	ln(ŷx)	(0, +∞)	e.g. x	ln sh x
(-∞, 0) or (0, +∞)	$\frac{1}{x^n}$ ($n = 2, 3, 4, \dots$)	$\frac{1}{1-\tilde{y}^n} \cdot \frac{1}{x^{n-1}}$	(-∞, 0)	e.g. x	ln sh (ŷx)
(0, +∞)	$x^{\tilde{y}}$ ($\tilde{y} \in \mathbb{R}, \tilde{y} \neq -1$)	$\frac{x^{\tilde{y}+1}}{\tilde{y}+1}$	R	$\frac{1}{\text{ch} 2x}$	th x
R	ex	ex	(-∞, 0) or (0, +∞)	$\frac{1}{\text{sh} 2x}$	$\tilde{y}$ for example x
(0, +∞)	ax ($a \in (0, +\infty), a \neq 1$)	$\frac{ax}{\ln a}$	R	$\frac{1}{1+x^2}$	facetg x = $\frac{\tilde{y}}{2} \tilde{y}$ facial x
R	sin x	$\tilde{y} \cos x$	(-1, 1)	$\frac{1}{1-\tilde{y}^2}$	arth x = $12^{-\ln 1 \frac{1+x}{\tilde{y}x}}$
R	cos x	sin x	(-∞, -1) or (1, +∞)	$\frac{1}{1-\tilde{y}^2}$	arch x = $12^{-\ln x \frac{x+1}{\tilde{y}1}}$
(-∞, -1), (-1, 1)	tg x	$\tilde{y} \ln \cos x$	R	$\frac{1}{\sqrt{1+x^2}}$	arsh x = $\ln(x + \sqrt{1+x^2})$
(0, ∞)	ctg x	ln sin x	(-1, 1)	$\frac{1}{\sqrt{1-\tilde{y}^2}}$	arcsin x = $\tilde{y}2 \tilde{y}$ arccos x
(-∞, -1), (-1, 1)	$\frac{1}{\cos^2 x}$	tg x	(1, +∞)	$\frac{1}{\sqrt{x^2-\tilde{y}1}}$	arch x = $\ln(x + \sqrt{x^2-\tilde{y}1})$
(0, ∞)	$\frac{1}{\sin^2 x}$	$\tilde{y} \text{ctg} x$	(-∞, -1)	$\frac{1}{\sqrt{x^2-\tilde{y}1}}$	arch (ŷx) = $\ln(\tilde{y}x + \sqrt{x^2-\tilde{y}1})$